🔑 **Answer Key — Teacher Copy · Lesson 2-4**

MSTAR PRACTICE WORKSHEET

Lesson 2-4: Represent and Describe Data in a Box Plot

Grade 6 Mathematics · Standard 6.DS.5

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

A box plot shows: Min = 10, Q1 = 15, Median = 20, Q3 = 28, Max = 35. What is the interquartile range (IQR)?

A. 10

B. 13

C. 20

D. 25

IQR = Q3 − Q1 = 28 − 15 = 13.

Question 2 SELECTED RESPONSE — CORRECT: A

On a box plot, what does the line inside the box represent?

A. The median

B. The mean

C. The mode

D. The range

The line inside the box of a box plot always represents the median.

Question 3 SELECTED RESPONSE — CORRECT: B

What percentage of data falls between Q1 and Q3 on a box plot?

A 25%

B 50%

C 75%

D 100%

> The box in a box plot contains the middle 50% of the data (from Q1 to Q3).

Question 4 SELECTED RESPONSE — CORRECT: C

Find the range of: 14, 22, 8, 31, 17

A 8

B 18

C 23

D 31

> Range = max − min = 31 − 8 = 23.

Question 5 SELECTED RESPONSE — CORRECT: A

Two box plots show: Team A has IQR = 8, Team B has IQR = 22. What does this tell you?

A Team A's middle 50% is more consistent; Team B's is more spread out

B Team B is a better team

C Team A has a higher median

D The IQRs don't tell you about consistency

> A smaller IQR means the middle 50% of data is more tightly clustered. Team A's scoring is more consistent in the middle range.

Question 6 SELECTED RESPONSE — CORRECT: B

The Hawks' points per quarter are 2, 4, 5, 7, 8, 10, 12. Resting on the scoreboard, what are the minimum, median, and maximum?

A Minimum 0, median 7, maximum 12

B Minimum 2, median 7, maximum 12

C Minimum 2, median 5, maximum 12

D Minimum 2, median 8, maximum 10

The data is already in order. The smallest value is 2 and the largest is 12. There are 7 values, so the median is the 4th value, which is 7.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

A box plot has a 5-number summary of minimum 4, Q1 = 8, median 12, Q3 = 18, maximum 24. What is the interquartile range?

A 6

B 10

C 12

D 20

IQR = Q3 − Q1 = 18 − 8 = 10.

- **A:** That is Q3 minus the median. The interquartile range spans Q1 to Q3.
- **C:** That is the median itself, which is a center rather than a spread. IQR = Q3 - Q1.
- **D:** That is the maximum minus the minimum, which is the range. IQR uses Q3 and Q1.

Part B — correct answer: C

What does the box itself represent in a box plot?

A All of the data from minimum to maximum.

B The mean of the data set.

C The middle 50% of the data, from Q1 to Q3.

D The values that occur most often.

Q1 marks the 25th percentile and Q3 the 75th, so the box spans the middle half of the data. The whiskers reach out to the minimum and maximum.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about box plots:

A The 5-number summary is minimum, Q1, median, Q3, and maximum.

B The box spans from Q1 to Q3.

C The line inside the box is the median.

D The whiskers show the mean of each half.

E IQR = Q3 − Q1.

A, B, C, and E are true. D is false — the whiskers extend to the minimum and maximum values, not to any mean.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Two basketball teams played 9 games each. Team A: 40, 42, 44, 46, 48, 50, 52, 54, 56. Team B: 30, 35, 40, 45, 48, 50, 55, 60, 65. Find the five-number summary and IQR for each team. Which team is more consistent? Which has a wider spread in the middle 50%?

Model answer: Team A's summary is 40, 43, 48, 53, 56 with IQR = 53 − 43 = 10, and Team B's is 30, 37.5, 48, 57.5, 65 with IQR = 57.5 − 37.5 = 20. Both teams share a median of 48, but Team B's middle 50% spreads twice as far, so Team A is the more consistent team.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
